GYEONGHUN GO

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EDUCATION

Chinese Academy of Sciences, Academy of Mathematics and Systems Science *2025.10-*
Postdoctoral researcher
Mentor: Yong Wang

POSTECH, Center for Mathematical Machine Learning and its Applications *2024.03-2025.08*
Research Fellow

Pohang University of Science and Technology *2018.02 - 2024.02*
Ph.D.
Department of Mathematics
Advisor: Donghyun Lee

Pohang University of Science and Technology *2013.03 - 2018.02*
B.S. in Mathematics

TALKS

Workshop on Fluid and Kinetic Partial Differential Equations Seminar *July 2026*
Title: Global Well-posedness for the Multi-species Boltzmann Equation with Large Amplitude Initial Data

The 12th Korea PDE Winter School *January 2026*
Title: Global Well-posedness for the Multi-species Boltzmann Equation with Large Amplitude Initial Data

Youngnam Mathematical Society-Chungcheong Mathematical Society Joint Annual Meeting *June 2025*
Title: Stability of the Boltzmann equation for a polyatomic gas with initial data allowing large oscillation

Yonsei University BK/BRL Seminar *June 2025*
Title: Dynamical billiard and a long-time behavior of the Boltzmann equation in general 3D toroidal domains

Korean Mathematical Society Presentation *April 2025*
Title: Global stability of the Boltzmann equation for a polyatomic gas with initial data allowing large oscillation

POSTECH-PNU Joint BK21 Mathematics Workshop *February 2025*
Title: Dynamical Billiard and a long-time behavior of the Boltzmann equation in general 3D toroidal domains

CM2LA Workshop *June 2024*
Title: Dynamical Billiard and a long-time behavior of the Boltzmann equation in general 3D toroidal domains

Webinar Kinetic and fluid equations for collective behaviors *June 2024*
Title: Dynamical Billiard and a long-time behavior of the Boltzmann equation in general 3D toroidal domains

KAIST, Partial Differential Equations Seminars *May 2024*
Title: Dynamical Billiard and a long-time behavior of the Boltzmann equation in general 3D toroidal domains

Hong Kong Society for Industrial and Applied Mathematcis *August 2023*
Title: Large-amplitude problem of the BGK model

International Congress on Industrial and Applied Mathematics 2023 Tokyo *August 2023*
Title: Large-amplitude problem of the BGK model

Korean Society for Industrial and Applied Mathematics Presentation *May 2023*
Title: Dynamical Billiard and a long-time behavior of the Boltzmann equation in general 3D toroidal domains

POSTECH BK workshop Presentation *January 2023*
Title: Large-amplitude problem of the BGK model

Korean Mathematical Society Presentation *October 2022*
Title: The Boltzmann equation in general solid torus domains

Korean Mathematical Society Presentation *October 2020*
Title: The Boltzmann equation with large-amplitude initial data in bounded domains

PUBLICATION AND PREPRINT

1. G.Ko, M.Lee, and S.Son, Global Well-posedness for the Multi-species Boltzmann Equation with Large Amplitude Initial Data, 2026

   Link: https://arxiv.org/pdf/2605.10138

2. G.Ko, S.Son, S. Cho, and M.Lee, A Theory-guided Weighted L^2 Loss for solving the BGK model via Physics-informed Neural Networks, 2026

   Link: https://arxiv.org/pdf/2604.04971

3. J.Kim and. G.Ko, Asymptotic behavior of large-amplitude solutions to the Boltzmann equation with soft interactions in $L^p_v L^\infty_x$ spaces, 2026

   Link: https://arxiv.org/pdf/2603.11903

4. G.Ko, C.Kim, and D.Lee, Dynamical Billiard and a long-time behavior of the Boltzmann equation in general 3D toroidal domains, *Tunisian Journal of Mathematics, 2025, Vol. 7, No. 2, 229–338.*

   Link: https://msp.org/tunis/2025/7-2/p01.xhtml

5. G.Ko and S.Son, Global Stability of the Boltzmann equation for a polyatomic gas with initial data allowing large oscillations, *J. Differ. Equ.*, 2025, Vol 425, pp. 506-552.

   Link: https://www.sciencedirect.com/science/article/pii/S0022039625000464

6. G.Bae, G.Ko, D.Lee, and S.Yun, Large amplitude problem of BGK model: Relaxation to quadratic nonlinearity, SIAM J. Math. Anal., 2026, Vol. 58, No. 2, pp. 1530-1570.

   Link: https://epubs.siam.org/doi/epdf/10.1137/25M1774537

7. G. Ko and D. Lee, On C^2 solution of the free transport equation in a disk, *Kinetic and Related Models*, 2023, 16(3): 311-372.

   Link: https://www.aimsciences.org/article/doi/10.3934/krm.2022031

8. G. Ko, D. Lee and K. Park, The large amplitude solution of the Boltzmann equation with soft potential. *J. Differ. Equ.*, 2022, Vol 307, pp. 297-347.

   Link: https://www.sciencedirect.com/science/article/pii/S0022039621006604

9. R. Duan, G. Ko and D. Lee, The Boltzmann equation with a class of large-amplitude initial data and specular reflection boundary condition, *Journal of Statistical Physics*, 2023, Vol 190, No. 189.

   Link: https://link.springer.com/article/10.1007/s10955-023-03203-6

TEACHING

Instructor

- **Introduction to Partial Differential Equations (MATH 313), POSTECH**

  February 2025 - June 2025

Teaching Assistant (Pohang University of Science and Technology)

- **Calculus I (MATH 101)**

  February 2023 - June 2023, February 2022 - June 2022, February 2021 - June 2021, February 2020 - June 2020, February 2018 - June 2018

- **Calculus II (MATH 102)**

  September 2021 - December 2021, September 2019 - December 2019, September 2018 - December 2018

- **Differential Equations (MATH 200)**

  September 2020-December 2020

- **Analysis I (MATH 311)**

  February 2019 - June 2019

- **Ordinary Differential Equations (MATH 412)**

  September 2022 - December 2022

Student Mentoring Program (Pohang University of Science of Technology)

- Calculus , Applied linear algebra, Real Analysis I (Mentor) *2015 - 2017*

SERVICE

Technical Research Personnel *February 2021 - February 2024*

HONORS/AWARDS

Beijing International Postdoctoral Exchange and Recruitment Program Funding *July 2026,Beijing Postdoctoral Research Foundation*

Excellent Dissertation Award *April 2024, Korean Mathematical Society*

Excellent Ph.D. Thesis Award *February 2024, POSTECH*

Graduate Research Fellowship *September 2023, POSTECH*

Postechian Fellowship (Innovation section) *November 2022, May 2022, POSTECH*

BK21 FOUR Excellent graduate students (Research) *November 2023, August 2022, POSTECH*

Excellent Teaching Assistant (Calculus II) *March 2022, February 2020, POSTECH*

Excellent Teaching Assistant (Calculus I) *October 2018, POSTECH*